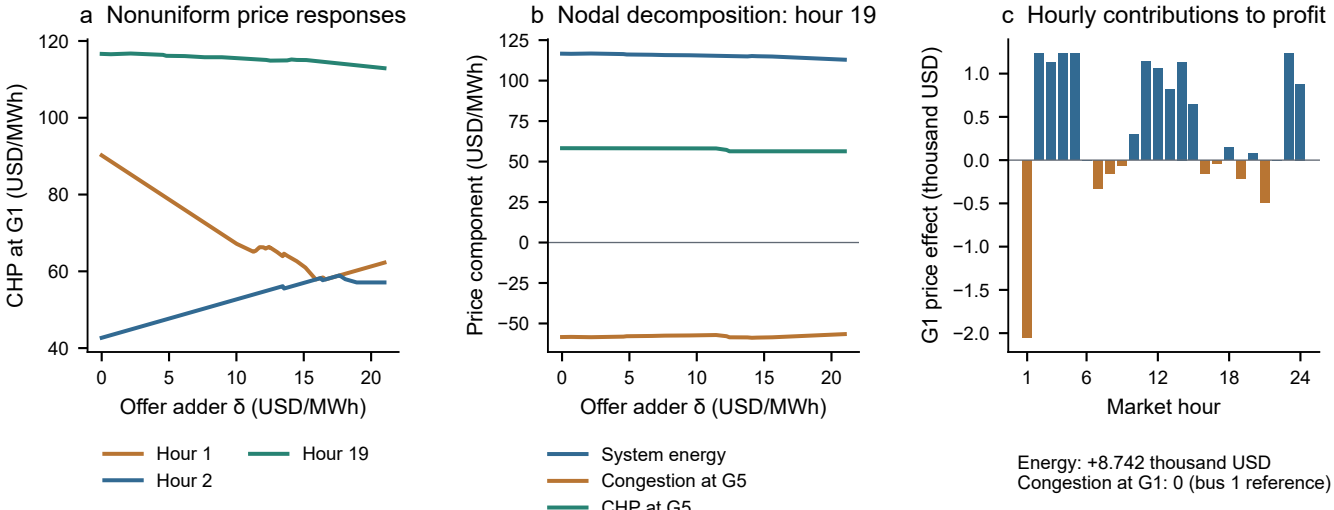
a Nonuniform price responses
CHP at G1 (USD/MWh)
120
100
80
60
40
0
5
10
15
20
Offer adder δ (USD/MWh)
Hour 1
Hour 19
Hour 2
b Nodal decomposition: hour 19
Price component (USD/MWh)
125
100
75
50
25
0
−25
−50
0
5
10
15
20
Offer adder δ (USD/MWh)
System energy
Congestion at G5
CHP at G5
c Hourly contributions to profit
G1 price effect (thousand USD)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
1
6
12
18
24
Market hour
Energy: +8.742 thousand USD
Congestion at G1: 0 (bus 1 reference)